

UNIT-III

ALGEBRIC STRUCTURE

→ GROUP:-

An Algebraic structure $(G, \circ)$ is said to be a group if it satisfies following Axioms.

1 CLOSURE AXIOM:-

If $a, b \in G$ then $a \circ b \in G$

2 IDENTITY AXIOM:-

If $a \in G$ then $e \in G$
s.t. $a \circ e = a \quad \forall a \in G$

3 INVERSE AXIOM:-

If $a \in G$ then element a^{-1} is called inverse of a if $a \circ a^{-1} = e$, where $a^{-1} \in G$.

4 ASSOCIATIVITY AXIOM:-

Let $a, b, c \in G$
 $a \circ (b \circ c) = (a \circ b) \circ c$

→ NOTE:-

1. An algebraic structure $(G, \circ)$ is called groupoid or quasi group if it satisfy only closure Axiom.
2. An algebraic structure $(G, \circ)$ is called semi-group or demi-group. if it satisfy closure and Associative Axioms only.
3. An Algebraic structure $(G, \circ)$ is called monoid if it is a Semi-group & also satisfy Identity Axiom.

→ ABELIAN GROUP OR COMMUTATIVE GROUP:

An algebraic structure $(G, \circ)$ is said to be abelian or commutative if it is a group & also satisfy Commutative Axiom
s.t $a, b \in G$, then $a \circ b = b \circ a$.

NOTE:-

An Algebraic structure $(G, \circ)$ is said to be finite if 'G' has finite no. of elements
otherwise, infinite.
No. of Elements is called order of 'G'.

Q:- Prove that the Set of cube root of unity is an Abelian group w.r to multiplication.

Solⁿ: We want to show that $(G, *)$ is an Abelian group where 'G' is the $G = \{1, \omega, \omega^2\}$

The Composition table of $(G, *)$ is

$*$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	$\omega^3 = 1$
ω^2	ω^2	$\omega^3 = 1$	$\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$ $= \omega$

→ CLOSURE AXIOM:-

In Composition table we see that the multiplication of each two elements lie in 'G'.

Hence, closure Axiom satisfy.

→ IDENTITY AXIOM:-

By Composition table it is clear that:

1 is the multiplicative Identity of each element.

i.e Identity Axiom satisfy.

→ INVERSE AXIOM:-

By Composition table it is clear that

- i) The Inverse of Element 1 is 1 .
 - ii) The Inverse of Element w is w^2 .
 - iii) The Inverse of Element w^2 is w .
- i.e Inverse Axiom is satisfy.

→ ASSOCIATING AXIOM:-

$$\begin{aligned} & 1 * (w * w^2) \\ &= 1 * w^3 \\ &= 1 * 1 \quad (\because w^3 = 1) \\ &= 1 \end{aligned}$$

Again,

$$\begin{aligned} & (1 * w) * w^2 \\ &= w * w^2 \\ &= w^3 \\ &= 1 \end{aligned}$$

$$\therefore 1 * (w * w^2) = (1 * w) * w^2$$

Hence, Associative Axiom is satisfy.

→ COMMUTATIVE AXIOM:-

In Composition table we see that Every two Elements of $(G, *)$ satisfy Commutative property.

Hence $G = \{1, w, w^2\}$ is an Abelian group.
w.r to multiplication.

Q:- Prove that the set $\{1, -1, i, -i\}$ is an Abelian group w.r to multiplication.

Sol:- The Composition table is -

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

1. CLOSURE AXIOM:-

In Composition table we see that the multiplication of Each two Elements lie in G .
Hence, Closure Axiom Satisfy.

2. IDENTITY AXIOM:-

By Composition table It is clear that 1 is the multiplicative Identity of each Element.

i.e Identity Axiom Satisfy.

3. INVERSE AXIOM:-

By Composition table It is clear that:

1. Inverse of Element 1 is 1.

2. Inverse of Element i is -i

3. Inverse of Element -1 is 1

4. Inverse of Element -i is i

i.e Inverse Axiom is Satisfy.

i is an
tion.

4 ASSOCIATIVE AXIOM:-

$$(1x-1)(ix-i)$$

$$\Rightarrow (-1)(-i^2)$$

$$\Rightarrow (-1)[-(-1)]$$

$$\Rightarrow (-1)(1)$$

$$\Rightarrow -1$$

$$[i^2 = -1]$$

Again;

$$(1xi)(-1x-i)$$

$$\Rightarrow (i)(i)$$

$$= i^2$$

$$\Rightarrow -1$$

$$\therefore (1x-1)(ix-i) = (1xi)(-1x-i)$$

Hence, Associative Axiom is Satisfy.

5 COMMUTATIVE AXIOM:-

that
ntity of

In Composition table we see that Every two Elements of $(G, *)$ satisfy Commutative property.

Hence, $G = \{1, -1, i, -i\}$ is an Abelian group w.r to multiplication.

that:

Q:- Show that the set $\{1, -1\}$ is an Abelian group w.r to multiplication.

Solⁿ:

$\times$	1	-1
1	1	-1
-1	-1	1

Solⁿ:

1. CLOSURE AXIOM:-

In Composition table we see that the multiplication of Each two Elements lie in G , Hence closure Axiom Satisfy.

2. IDENTITY AXIOM:-

By Composition table It is clear that 1 is the Multiplicative Identity of Each Element.

3. INVERSE AXIOM:-

- 1 Inverse of Element 1 is 1
- 2 Inverse of Element -1 is 1

4. ASSOCIATIVE AXIOM:-

(1×-1)	(-1×1)	(-1×1)	(1×-1)
$(-1)(-1)$		$(-1)(-1)$	
1		1	

5. COMMUTATIVE AXIOM:-

In Composition table it is clear that Commutative property Satisfy.

Q:- Show that the Set, $A = \{a, b, c\}$ forms an Abelian group of order 3. w.r to multiplication acc. to following composition table.

Solⁿ:-

$*$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

1. CLOSURE AXIOM:-

In composition table we see that the multiplicative of each two elements lie in 'G', hence closure axiom satisfy.

2. IDENTITY AXIOM:-

By composition table it is clear that 'a' is the multiplicative identity for each element.

3. ASSOCIATIVE AXIOM:-

$(a * b) * c$	$a * (b * c)$
$b * c$	$a * a$
a	a

4. INVERSE AXIOM:-

- 1. Inverse of Element a is a
- 2. Inverse of Element b is c
- 3. Inverse of Element c is b

5. COMMUTATIVE AXIOM:-

In composition table it is clear that commutative property satisfy.

Q:- Show that the set of all Integers is an ∞ Abelian group w.r to Addition.

Sol:- Let $(G, +)$ is an Algebraic structure
 = Set $G = \{-\infty, \dots, -2, -1, 0, 1, 2, \dots, \infty\}$

1. Closure Axiom -

Since Addition of two Integers is also an Integer i.e closure axiom satisfy.

2. Identity Axiom -

Zero is the Additive identity for each Element of 'G' & also is an Integer.

3. Inverse Axiom -

In set 'G' Additive Inverse of each element also Exist in 'G'.

4. Associative Axiom -

Integers good hold Associativity w.r to addition.

5. Commutativity Axiom -

Integers good hold Commutativity w.r to addition.

$\therefore$ Since 'G' has ∞ Elements & satisfy above properties Hence 'G' is a ∞ Abelian group w.r to Addition.

Q1- Show that the set of all Even Integers including zero is also an Abelian group w.r to Addition.

Soln- Let $(G, +)$ is an Algebraic structure.
 s.t $G = \{0, \pm 2, \pm 4, \pm 6, \pm 8, \dots, \pm \infty\}$

1 Closure Axiom-

The sum of Every two Integers is also Even integer.

2 Identity Axiom:-

Zero is the Additive Identity for Every Element.

3 Inverse Axiom:-

Every Element of Set 'G' has their additive inverse.

4 Associative Axiom:-

Even Integers good hold Associativity w.r. to Addition.

5 Commutativity Axiom:-

Even Integers good hold Associativity.

$\therefore$ Since G has ∞ Elements & satisfy all properties, Hence G is a abelian grp w.r to addition.

Q.1 Show that the Set of all non-zero Complex no.s is a multiplicative group.

Solⁿ let $(G, *)$ is an Algebraic structure.
s.t $G = \{a+ib; a, b \in \mathbb{R}\}$

1. Closure Axiom:-

let $a+ib, c+id \in G$ s.t $a, b, c, d \in \mathbb{R}$
Now,

$$\begin{aligned} (a+ib) * (c+id) \\ ac + iad + ibc + i^2 bd \\ (ac - bd) + i(ad + bc) \end{aligned}$$

i.e product is also a complex no.

2. Identity Axiom:-

$(1+i0)$ is the multiplicative identity for Each Complex no. of set G .

3. Additive Axiom:-

let $a+ib$ is the multiplicative Inverse of $c+id$.

$$(a+ib) * (c+id) = 1+i0$$

$$a+ib = \frac{1+i0}{c+id}$$

$$= \frac{1}{c+id} \times \frac{c-id}{c-id}$$

$$= \frac{c-id}{c^2+d^2} \rightarrow \left(\frac{c}{c^2+d^2} \right) + i \left(\frac{-d}{c^2+d^2} \right)$$

i.e multiplicative Inverse of Each Complex no. Exist.

4. Associativity:-

We know that, Complex no.s good hold Associativity w.r to multiplication.

Hence the set of all non-zero complex no. is a multiplicative grp.

Q:-

Prove that the set of matrices.

$$A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \text{ where } \alpha \text{ is a Real}$$

is a multiplicative group.

Solⁿ - Closure -

$$= A_\alpha \cdot A_\beta$$

$$\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}$$

$$\begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

$$\because \alpha, \beta \in \mathbb{R} \Rightarrow \alpha + \beta \in \mathbb{R}.$$

closure Axiom Satisfy.

2
=

Identity -

$$\text{At } \alpha = 0$$

$$A_0 = \begin{bmatrix} \cos 0 & \sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is multiplicative Identity.

3 Inverse Axiom :-

A_α is a Non-singular matrix $\forall \alpha \in \mathbb{R}$
i.e. multiplicative Inverse will Exist.

4 Associative Axiom :-

$$\therefore A_\alpha \cdot A_\beta = \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

and

$$(A_\alpha \cdot A_\beta) \cdot A_\gamma = \begin{bmatrix} \cos\{(\alpha + \beta) + \gamma\} & -\sin\{(\alpha + \beta) + \gamma\} \\ \sin\{(\alpha + \beta) + \gamma\} & \cos\{(\alpha + \beta) + \gamma\} \end{bmatrix}$$

$\therefore$ Real no.s good hold Associativity w.
r. to Addition.

$$\text{i.e. } (\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$$

Hence set of matrices is a multiplicative group.

Q:- If Q^+ be the set of all +ve Rational no's
 & $*$ be a Binary Composition defined by
 $a * b = \frac{ab}{3}$, $a, b \in Q^+$

Show that $(Q^+, +)$ is a commutative group.
 find the identity of the group.

Solⁿ: Relation -

! Closure $a * b = \frac{ab}{3}$ Satisfy closure Axiom $\forall Q^+$

Eg:- Let $a = \frac{1}{2}$, $b = \frac{3}{5}$

$$a * b = \frac{ab}{3}$$

$$\frac{1}{2} * \frac{3}{5} = \frac{\frac{1}{2} * \frac{3}{5}}{3}$$

$$= \frac{1}{10} \in Q^+ \text{ which is also a +ve Relation no.}$$

2 Identity:-

Let 'a' be the identity of 'b'

$$\text{i.e. } a * b = b$$

$$\frac{ab}{3} = b$$

$$a = 3$$

Hence '3' is the identity for Every Rational no.

3

Inverse -

let 'c' is the Inverse of 'd'

i.e

$$c \times d = 3$$

$$\Rightarrow \frac{cd}{3} = 3$$

$$\Rightarrow c = 9/d$$

i.e Every +ve Rational no. has there Inverse.

$$\text{Eg:- } \text{inv}\left(\frac{2}{5}\right) = \frac{9}{2/5} = 45/2$$

4

Associativity:-

$$a \times b = \frac{ab}{3}$$

$$a \times b = \frac{ba}{3}$$

 $\therefore$ Associative Axiom satisfy.

Imp

Q:- Prove that the set $G = \{0, 1, 2, 3, 4\}$ is a finite Abelian group of order 5 w.r to Addition under modulo 5,

Sol^m:- The Composition table under modulo 5 is

$+5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

1. Closure:-

In Composition table we see that addition of Each two Elements is possible & Resultant Exist in G .

2. Identity:-

0 is the additive Identity of Each Element
 $\therefore$ Identity Axiom Satisfy.

3. Inverse Axiom:-

The Additive Identity of 0 is 0.

The Additive Identity of 1 is 4.

" " " " 2 is 3

" " " " 3 is 2

" " " " 4 is 1

4 Associativity:-

$$(1+2)+3 = 1+(2+3)$$

$$\begin{array}{ccc} 3+3 & = & 1+0 \\ 1 & = & 0 \end{array}$$

∴ Associative Axiom Satisfy.

5 Commutative Axiom

In Composition table we see that Every two Elements satisfy Comm.

∴ $\{G, +\}$ is an Additive Abelian group of order 5.

→ SUB GROUP -

A Subset of a grp G is called Sub Group. It is also a grp on inducing The Composition of grp.

Eg:- Let $(G, \circ)$ is a group st $H \subseteq G$, then H is called Subgroup of G if $(H, \circ)$ is also a group.

→ THEOREM -

The necessary & sufficient Conditions for a Subset of a group to be a Sub group are;

1) A Subset H of a group G is a subgroup iff

i) $a \in H, b \in H \Rightarrow a \circ b \in H$

ii) $a \in H \Rightarrow a^{-1} \in H$

Proof:- i) Let H is a subgroup of group G

if $a, b \in H$

$\Rightarrow a \circ b \in H \subseteq G$

$\Rightarrow a \circ b \in H$

ii) Let $a \in H$

$\because a \in H \subseteq G$

$\Rightarrow a \in G$

$\Rightarrow a^{-1}$ exist

Converse -

Given that -

i) $\rightarrow a \in H, b \in H \Rightarrow a \circ b \in H \rightarrow \textcircled{1}$

ii) $\rightarrow a \in H \Rightarrow a^{-1} \in H \rightarrow \textcircled{2}$

Now we want to show that H is also a group.

i) Closure Axiom -

By (1)

$a \in H, b \in H \Rightarrow a \circ b \in H$

which is a closure Axiom.

→ Inverse Axiom -

By (iv)

$a \in H \Rightarrow a^{-1} \in H$ which is also an Inverse Axiom.

→ Identity Axiom -
by (v)

$$a \in H \Rightarrow a^{-1} \in H$$

$$\Rightarrow a o a^{-1} \in H \quad \text{by (i)}$$

$$\Rightarrow e \in H$$

$\therefore e$ exist in H .

→ Associativity Axiom -
by (v)

$$a, b, c \in H \Rightarrow a o b \in H \subseteq G \text{ and } c \in H \subseteq G$$

$$\Rightarrow a o b \in H \text{ and } c \in H \Rightarrow (a o b) o c \in H$$

Again,

$$\Rightarrow (a o b) o c \in G \quad \because H \subseteq G$$

$$c, b, a \in H \Rightarrow b o a \in H \subseteq G \quad \text{--- (ii)}$$

$$\Rightarrow b o a \in G$$

$$a \in H \text{ and } b, c \in H$$

$$\Rightarrow a \in H \text{ and } (b o c) \in H \text{ by (i)}$$

$$\Rightarrow a o (b o c) \in H$$

$$\Rightarrow a o (b o c) \in G \quad \because H \subseteq G$$

But in G

$$(a o b) o c = a o (b o c)$$

Hence, Associative Axiom Satisfy.

→ THEOREM -

The intersection of two subsets of a group G is a sub group of G .

proof:- Let H_1 & H_2 are two subgroups of a group G
then, $H_1 \cap H_2 \neq \emptyset$

let $a, b \in H_1 \cap H_2$

$\Rightarrow a \in H_1, \cap H_2$ and $b \in H_1, \cap H_2$

~~$\Rightarrow a \in H_1, \cap H_2$ and $(b \in H_1 \text{ and } b \in H_2)$~~

~~$\Rightarrow a \in H_1, \cap H_2$ and~~

$\Rightarrow a, b \in H_1$ and $a, b \in H_2$

$\Rightarrow ab^{-1} \in H_1$ and $ab^{-1} \in H_2$

$\Rightarrow ab^{-1} \in H_1 \cap H_2$ — ①

i.e

If $a \in H_1 \cap H_2, b \in H_1 \cap H_2$

$\Rightarrow ab^{-1} \in H_1 \cap H_2$

which is required condition for $H_1 \cap H_2$ to be a sub group.

(Necessary & sufficient condition for H_1, H_2 to be ^{sub}group)

→ THEOREM -

A necessary and sufficient condition for a non-Empty subset H of a group G to be a Sub-group is that $a \in H, b \in H \Rightarrow ab^{-1} \in H$, where b^{-1} is the Inverse of b .

Proof:- If Part -

Let H be a sub group s.t $a, b \in H$

$$\because a, b \in H$$

$$\Rightarrow a \in H, b \in H$$

$$\Rightarrow a \in H, b^{-1} \in H \quad (\text{Inverse axiom in } H)$$

$$\Rightarrow ab^{-1} \in H \quad (\text{By closure Axiom})$$

only If Part:-

Now we have $a \in H, b \in H \Rightarrow ab^{-1} \in H$.

and we want to show that H is a subgroup.

→ Closure axiom -

$$\because ab^{-1} \in H \quad \forall a \in H, b \in H$$

$$\Rightarrow a \in H \text{ \& } b^{-1} \in H \quad (\text{By given Cond})$$

$$\text{i.e. if } a \in H \text{ \& } b^{-1} \in H \Rightarrow ab^{-1} \in H$$

which is closure Axiom.

→ Inverse Axiom:-

$$\because b \in H \text{ s.t } b^{-1} \in H.$$

Inverse Axiom Satisfy.

→ Identity Axiom:-

$$\because b^{-1} \in H \quad \forall b \in H$$

$$\Rightarrow b^{-1}b \in H \quad (\text{by closure Axiom})$$

$\Rightarrow e \in H \quad \forall b \in H$, Identity Axiom Satisfy.

→ Associative Axiom:-

All ~~Axiom~~ Elements of Subset H are the Elements of group G , and all Elements of G satisfy associativity.

i.e Elements of subset H satisfy Associativity.

Hence, $a \in H, b \in H \Rightarrow ab \in H$ is necessary & sufficient condition to make H a Subgroup.

→ THEOREM:-

A finite Subset H of a group G is a subgroup of G , iff $a, b \in H \Rightarrow ab \in H$.

proof:- Condition is necessary -

$\therefore H$ is a Subgroup

i.e. $a, b \in H \Rightarrow ab \in H$ (by closure Axiom)

Condition is sufficient:-

Now we have $a, b \in H \Rightarrow ab \in H$

and we want to show H is a Subgroup of G .

i) Closure Axiom -

Given that,

$a, b \in H \Rightarrow ab \in H$ which is closure Axiom.

ii) Associative Axiom -

let $a, b, c \in H$

$\Rightarrow a, b, c \in H \subseteq G$

$\Rightarrow a, b, c \in G$

$\therefore G$ is a group i.e its all Elements satisfy associativity.

iii) Identity Axiom:-

let $a \in H$

$\Rightarrow a \in H, a \in H \Rightarrow a \cdot a \in H$ (By closure)

$\Rightarrow a^2 \in H$

Again,

$a^2 \in H, a \in H \Rightarrow a^3 \in H$ (By closure)

~~i.e~~

i.e

$$\text{If } a \in H$$

$$\Rightarrow a, a^2, a^3, \dots, a^n \in H. \quad \text{--- (1)}$$

i.e H contains ∞ Elements, which is Contradiction.

B'coz Given that H is finite.

i.e In Relation (1) Some Elements will be Repeated.

$$\text{let } a^r = a^s$$

$$\Rightarrow a^r \cdot a^{-s} = a^s \cdot a^{-s}$$

$$\Rightarrow a^{r-s} = e \quad \text{where } r-s \geq 1$$

i.e Identity exist.

IV) Inverse Axiom -

$$\because r-s \geq 1$$

$$\Rightarrow r-s-1 \geq 0$$

i.e

$$a^{r-s-1} \in H$$

$$\Rightarrow a^{r-s} a^{-1} \in H$$

$$\Rightarrow e \cdot a^{-1} \in H$$

$$\Rightarrow a^{-1} \in H$$

$$\text{If } a \in H \Rightarrow a^{-1} \in H.$$

→ Cyclic Group:-

A group $(G, \circ)$ is called cyclic if, for some element $a \in G$, Every Element $x \in G$ is of the form a^n , where n is some integers a called generator of G .

→ THEOREM -

Every cyclic group is Abelian.

Proof:- let $(G, *)$ be a grp s.t a is the generator of G
 of $a^r, a^s \in G$ where $r, s \in \mathbb{I}$
 Now

$$\begin{aligned}
 a^r \cdot a^s &= a^{r+s} \\
 &= a^{s+r} \quad (\text{Additive Comm. of Integers}) \\
 &= a^s \cdot a^r
 \end{aligned}$$

i.e group is Abelian.

Hence, Every cyclic grp is Abelian.

→ THEOREM -

The order of a cyclic group is same as the order of its generator.

Proof:- let $(G, *)$ be a Abelian group s.t a is its generator with order n .
 $\because a \in G$ s.t $o(a) = n$
 i.e $a^n = e$

let s be the order of grp.

$$0 \leq s \leq n$$

$$\text{s.t } n = sq + r$$

$$\therefore a^n = e$$

$$a^{sq+r} = e$$

$$\Rightarrow a^{sq} \cdot a^r = e$$

$$\Rightarrow (a^s)^q \cdot a^r = e$$

$$\Rightarrow (e)^q \cdot a^r = e$$

$$\Rightarrow e \cdot a^r = e$$

$$\Rightarrow a^r = e$$

which is not true

$$\therefore a^n = e$$

Converse -

Again if -

$$0 \leq n \leq s$$

$$\text{i.e. } s = qn + r$$

$$a^s = a^{qn+r}$$

$$= (a^n)^q \cdot a^r$$

$$= e^q \cdot a^r$$

$$= a^r$$

but if $a^s = e$

$$\text{i.e. } a^r = e$$

which is Again not possible.

Hence,

The order of a group & the generator is same.

OK ✓

→ COSETS:-

If G is a group, H is a subgroup of G
s.t. $h \in H$ and $a \in G$, then $\{ha : h \in H\}$
is called Right Coset.

Similarly; $\{ah : h \in H\}$ is called left Coset.

Eg:- If H, K are two subgroups of a group G
iff $HK = KH$.

Solⁿ:- let we have $HK = KH$

now we wish to show that H and K are two
subgroups.

$$\text{i.e. } (HK)(HK)^{-1} = HK \quad [\because HH^{-1} = H]$$

[required condition to be a subgroup]

$$\text{if } (HK)(HK)^{-1} = HK$$

$$\Leftrightarrow (HK)(K^{-1}H^{-1}) = HK$$

$$\Leftrightarrow H(KK^{-1})H^{-1} = HK$$

$$\Leftrightarrow HKH^{-1} = HK$$

$$\Leftrightarrow (KH)H^{-1} = HK$$

$$\Leftrightarrow K(HH^{-1}) = HK$$

$$\Leftrightarrow KH = HK$$

Hence proved.

Q:- $G = \{a, a^2, a^3, a^4 = 1\}$, $H = \{1, a^2\}$ is a
subgroup of G . Find all Cosets of H in G .

and prove that G is Equal to the
union of all these Cosets and also Establish
that any two cosets are either disjoint or
identical.

Solⁿ:- i) Right Cosets -

$$Ha = \{1, a^2\} \times a = \{a, a^3\}$$

$$Ha^2 = \{a^2, 1\}$$

$$Ha^3 = \{a^3, a\}$$

$$Ha^4 = \{1, a^2\}$$

Left Cosets -

$$aH = \{a, a^3\}$$

$$a^2H = \{a^2, 1\}$$

$$a^3H = \{a^3, a\}$$

$$a^4H = \{1, a^2\}$$

$$ii) Ha \cup Ha^2 \cup Ha^3 \cup Ha^4$$

$$= \{a, a^3, a^2, 1\}$$

$$= \{1, a, a^2, a^3\}$$

$$= G$$

Similarly;

$$aH \cup a^2H \cup a^3H \cup a^4H$$

$$= \{a, a^3, a^2, 1\}$$

$$= \{1, a, a^2, a^3\}$$

$$= G$$

iii) We See that

$$Ha = aH = Ha^3 = a^3H = \{a, a^3\} \text{ --- (1)}$$

and

$$Ha^2 = a^2H = Ha^4 = a^4H = \{1, a^2\} \text{ --- (2)}$$

By (1) & (2) We Conclude that;

any two cosets are either disjoint or identical.

→ THEOREM :-

If $h \in H$, then the Coset hH (or Hh) is identical with H , and Conversely.

Soln:-

let $h \in H$ and $h^{-1} \in H$.

$$\Rightarrow hh^{-1} \in H \text{ --- (I)}$$

$\therefore H$ is a subgroup

$\therefore h, h' \in H$.

$$hh' \in H \text{ --- (II) (by closure Axiom)}$$

by (I) & (II)

$$hH \subseteq H \text{ --- (A)}$$

but we know that every Coset is the Superset of that set (group)

$$\text{i.e. } H \subseteq hH \text{ --- (B)}$$

By (A) & (B)

$$hH = H$$

→ THEOREM

Any two cosets (left or right) of H are either disjoint or ^{to} be identical.

proof:- let aH and bH are two Cosets of H .
s.t these are not identical disjoint.

Therefore, let

$$aH \cap bH = c \text{ (say)}$$

$$\text{s.t } ah = bh' = c$$

$$\therefore ah = bh'$$

$$\Rightarrow (ah)h^{-1} = (bh')h^{-1}$$

$$\Rightarrow a(hh^{-1}) = (bh')h^{-1}$$

$$\Rightarrow a = (bh')h^{-1}$$

$$\Rightarrow a = b(h'h^{-1}) \text{ --- (I)}$$

$\because h' \in H \cdot h' \in H \Rightarrow h'h^{-1} \in H$ (by closure)
 let $h'h^{-1} = h''$ (say)

by (1)

$$a = bh''$$

$$\Rightarrow aH = (bh'')H$$

$$\Rightarrow aH = b(h''H)$$

$$\Rightarrow aH = bH \quad (\because hH = H)$$

i.e If we choose two non-disjoint cosets, then they will be identical.

→ THEOREM -

If $aH = bH$ where $a, b \in G$, then $ab^{-1} \in H$

Also show that Converse is true.

proof:-

$$\text{let } aH = bH$$

then

$$\because ah = bh' \quad \text{where } h \in H$$

$$\Rightarrow (ah)h^{-1} = (bh')h^{-1}$$

$$\Rightarrow a(hh^{-1}) = (bh'h^{-1})$$

$$\Rightarrow a = (bh'h^{-1})$$

$$\Rightarrow a = b(h'h^{-1}) \quad \text{--- (1)}$$

$$\because h' \in H \cdot h' \in H \Rightarrow h'h^{-1} \in H \text{ (by closure)}$$

$$\text{let } h'h^{-1} = h'' \text{ (say)}$$

by (1)

$$a = bh''$$

$$ab^{-1} = (bh'')b^{-1}$$

$$= b(h''b^{-1})$$

$$= (bb^{-1})h''$$

$$= eh''$$

$$= h''$$

$$\circ \circ \quad h^{-1} \in H \Rightarrow ab^{-1} \in H$$

Converse

$$\text{let } ab^{-1} \in H$$

$$\text{let } ab^{-1} = h \text{ (say)}$$

$$\Rightarrow a = bh$$

$$\Rightarrow aH = (bh)H$$

$$\Rightarrow aH = b(hH)$$

$$\Rightarrow aH = bH$$

$\Rightarrow$

THEOREM -

If $a, b \in G$ s.t. $a \neq b$ and H is subgroup of G ,
then

$$a \in bH \Leftrightarrow aH = bH$$

$$\text{and } a \in Hb \Leftrightarrow Ha = Hb$$

proof:-

$$\text{let } a \in bH$$

$$\Rightarrow ab^{-1} \in (bH)b^{-1}$$

$$= b(Hb^{-1})$$

$$= b(b^{-1}H)$$

$$= (bb^{-1})H$$

$$= eH$$

$$= H$$

$$\Rightarrow ab^{-1} \in H$$

$$\Rightarrow (ab^{-1})H = HH$$

$$\Rightarrow (ab^{-1})H = H$$

$$\Rightarrow b(ab^{-1})H = bH$$

$$\Rightarrow (ab^{-1})bH = bH$$

$$\Rightarrow a(b^{-1}b)H = bH$$

$$\Rightarrow aH = bH$$

Converse,

$$\text{let } aH = bH$$

$$\text{s.t. } a \in aH \Rightarrow a \in bH$$

proved

i) $a \in Hb \Rightarrow Ha = Hb$

let $a \in Hb$

$\Rightarrow ab^{-1} \in (bH)b^{-1}$

$\Rightarrow = (Hb)b^{-1}$

$= H(bb^{-1})$

$= He$

$= H$

$\Rightarrow ab^{-1} \in H$

$H(ab^{-1}) \subseteq HH$

$\Rightarrow H(ab^{-1}) = H$

$\Rightarrow H(ab^{-1})b = Hb$

$\Rightarrow Ha(b^{-1}b) = Hb$

$\Rightarrow Ha = Hb$

Converse,

let $Ha = Hb$

So $a \in Ha \Rightarrow a \in Hb$

H.M. Jwp

THEOREM-

State and prove Lagrange's theorem.

State -

The order of a subgroup of a finite group is a divisor of the order of a group.

proof:- let G is a group of order n and H be a subgroup of G with order m .

So

$H = \{h_1, h_2, h_3, \dots, h_m\}$

If $a \in G$ then

$aH = \{ah_1, ah_2, \dots, ah_m\}$

if it is possible that

$$ah_i = ah_j$$

$$h_i = h_j$$

[by Cancellation law]

which is not possible

i.e. order of each coset will be m .

$\because G$ has n distinct elements ($\because$ order is n)

i.e.

$$o(G) = o(aH) \times \text{no. of cosets}$$

$$o(G) = o(aH) \times k \text{ (say)}$$

$$n = m \times k$$

$$k = \frac{n}{m}$$

i.e. $\frac{\text{order of finite grp}}{\text{order of subgroup}} = \text{No. of cosets}$

i.e. m is divisible by n } $\downarrow$

→ THEOREM -

If H and K are finite subgroups of a group G , then

$$\frac{o(HK)}{o(H \cap K)} = \frac{o(H) o(K)}{o(H \cap K)}$$

proof - Given that H and K are finite subgroups of a group G ,

Here $o(HK) = \text{No. of distinct Element in } (HK)$

let

$$D = H \cap K$$

where D will be a subgroup of G . s.t. $D \subseteq K$

i.e. D will be a subgroup of K .

($\because K$ is a subgroup)

Hence,

$$\frac{o(K)}{o(D)} = n \text{ (say)} \quad \text{(by Lagrange's theorem)} \quad \textcircled{ii}$$

Again,

$$K = DK_1 \cup DK_2 \cup DK_3 \cup \dots \cup DK_n$$

$$K = \bigcup_{i=1}^n DK_i \quad (DK_i \text{ are right cosets})$$

$$\Rightarrow HK = \bigcup_{i=1}^n H \cup DK_i$$

$$= \bigcup_{i=1}^n HK_i$$

$$= \bigcup_{i=1}^n HK_i$$

$$\because HD = H$$

$$\because D \subseteq H$$

$$\Rightarrow HK = HK_1 \cup HK_2 \cup \dots \cup HK_n$$

If it possible

$$HK_i = HK_j$$

$K_i = K_j$ which is not possible.

i.e

$$o(HK) = \text{no. of cosets} \times \text{order of each coset}$$

$$= n \times o(H)$$

$$= \frac{o(K)}{o(D)} \times o(H)$$

$$= \frac{o(H) \cdot o(K)}{o(D)}$$

Hence

$$= \frac{o(H) \cdot o(K)}{o(D)}$$

$$(\because D = H \cap K)$$

$$\frac{o(K)}{o(D)} = n \times \frac{o(H \cap K)}{o(D)}$$

$\nexists$